

Optimizing UAV Jammer for Covert Communication of LEO Satellite System

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Abstract—With the development of satellite communications, their security issues have received widespread attention. We investigate the covert performance of the low earth orbit (LEO) satellite communication system in this paper, where the unmanned aerial vehicle (UAV) is regarded as a jammer. For practical consideration, we assume that the UAV can only obtain part of the environmental information, and the detection threshold of the eavesdropper on the ground is not known. It derives the covert rate and the probability of detection errors of the eavesdropper. Furthermore, the genetic algorithm (GA) is proposed to achieve the best covert rate by optimizing the power and trajectory scheme of UAV. The simulation results show that the UAV jamming solution generated by GA can effectively improve the covert performance of the LEO satellite communication system.

Keywords—LEO satellite; covert communication; genetic algorithms;

I. INTRODUCTION

The use of satellite communication, which can offer extensive coverage, high communication capacity, and practical realization services, is seen as a promising technology for future global personal communication [1]. Due to their low power consumption, low latency, and flexible deployment capabilities, low earth orbit (LEO) satellites with orbital altitudes of less than 1500 km in particular have seen significant development in recent years. However, due of the large transmission distance and open wireless channels, LEO satellite communication signals are severely intercepted.

The security of satellite communication is mainly achieved by upper-level encryption technology. However, due to the increasement of the computing power of eavesdroppers, traditional encryption technology is difficult to meet the requirements of covert communication. Physical layer security (PLS) strategies were recommended as a

way to implement the security of upper layers by taking advantage of the physical characteristics of wireless media [2]. Although PLS is committed to ensuring that eavesdroppers can't successfully demodulate the signal, it still can't guarantee the undetectability of the signal. Once the eavesdropper observes the communication behavior, the signal and the location of the legitimate node will be exposed. In practical Wireless communication systems, there are various forms of jamming, interception and detection in the space environment, if the location of the ground terminal can't be detected by eavesdroppers, the communication system can achieve relatively higher security performance. LEO Satellite Communications is therefore prepared to pursue covert communication at a higher level of security.

The purpose of covert communication, commonly referred to as low likelihood of detection (LPD) communication, is to make wireless transmissions undetectable. Based on [3], covert communication can be improved by using power control, noise uncertainty, channel uncertainty, etc. However, it is challenging to completely mask the communication signal in the background noise in real-world wireless communication scenarios. As an effective method, cooperative artificial noise (AN) has been used to assist covert transmission [4], where most of the cooperative nodes are mainly static.

With the advantages of flexible deployment and easy control, unmanned aerial vehicles (UAVs) can initiate AN to effectively confuse the detection of the eavesdropper, ensuring security transmission of information by controlling its operation of approaching the eavesdropper and sending interference noise [5–7]. As the mobile jamming node, UAVs' trajectory and interference power are optimized by different algorithms, such as successive convex approximation (SCA) [5], block coordinate descent based iterative algorithm [6],

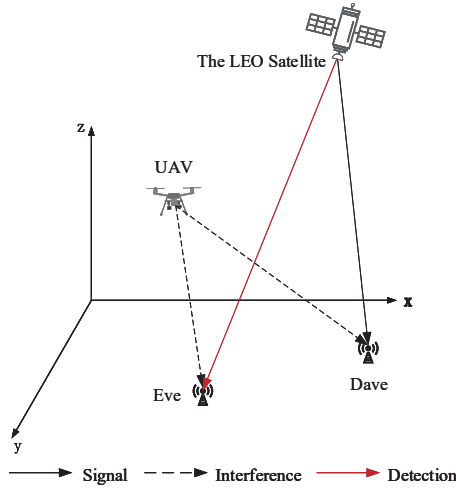


Figure 1. System Model

Dinkelbach method [7], etc. These proposed optimization algorithms, however, are based on the unrealistic assumption that UAVs have perfect channel distribution information (CDI) and the eavesdropper detection threshold because there is no signal exchange between them. Furthermore, it is challenging to apply these algorithms to more difficult non-convex optimization problems, they can only be used to solve convex optimization problems. Therefore, in this paper, the UAVs obeying a 3D trajectory are applied to achieve covert communication of LEO satellites through cooperative jamming, where the partial information of eavesdropper is considered.

In the satellite communication system presented herein and based on the case of UAV unknown perfect CDI and detection thresholds, we intend to get the optimal trajectory and jamming power for the UAV by using Genetic Algorithms (GA) to maximize the security performance of the system. The simulation results show the effectiveness of UAV's jamming solution which is generated by using GA. In contrast to other benchmark algorithms, GA can jump out of the local optimal solution and obtain the global optimal solution with only a partial understanding of CDI.

II. SYSTEM MODEL

We investigate covert transmission for a downlink low earth orbit (LEO) satellite communication system, as depicted in Fig.1. In particular, the terrestrial legal user (Dave), is receiving private information against the eavesdropping of Eve. Additionally, a cooperative UAV is used to dynamically generate jamming signal to hinder the eavesdropper's detection in order to aid the LEO satellite's secret communication.

We assume, without sacrificing generality, that the covert communication takes place over a short period of time T , which can be separated into N consecutive time slots, the time of which is denoted by $\delta_t = \frac{T}{N}$. The positions of all communication nodes are placed in a Cartesian three-

dimensional (3D) coordinate, in order to characterize the Euclidean distance between any two nodes. Consequently, the UAV's position in the n th time slot can be represented as $(x_U[n], y_U[n], z_U[n])$, $n \in \{1, 2, 3 \dots N\}$. We suppose that $\omega_0[n]$ and $\omega_1[n]$, respectively, represent the angles between the UAV's flying direction and the positive horizontal/vertical direction. In addition, the UAV's take-off position is predetermined, then the UAV fly as the speed $v[n]$ for straight flight with its position in n th ($n \in \{1, \dots, N\}$) time slot expressed as

$$\begin{aligned} (x_U[n] + v[n]\delta_t \cos \omega_0[n], y_U[n] + v[n]\delta_t \sin \omega_0[n], \\ z_U[n] + v[n]\delta_t \cos \omega_1[n]). \end{aligned} \quad (1)$$

Furthermore, Dave stays static during the whole communication period, and its location is denoted by $(x_D, y_D, 0)$. The distance between the LEO satellite and Dave is denoted by d_{SD} . Hence, the received signal power of Dave is $P_D[n] = \frac{P_S[n]}{d_{SD}^\alpha}$, where $P_S[n]$ stands for the LEO satellite's transmit power and α denotes the path-loss exponent. While the distance between Dave and the UAV, d_{UD} , varies during the UAV's flight period, which is expressed as

$$d_{UD}[n] = \sqrt{(x_U[n] - x_D)^2 + (y_U[n] - y_D)^2 + (z_U[n])^2}. \quad (2)$$

Additionally, let's assume that the legitimate nodes are aware of Eve's precise location, which allows us to dynamically describe the distance between UAV and Eve as

$$d_{UE}[n] = \sqrt{(x_U[n] - x_E)^2 + (y_U[n] - y_E)^2 + (z_U[n])^2}. \quad (3)$$

Next, it is assumed that the system model's nodes are each outfitted with a single antenna. The channel between the LEO satellite and Dave is the Shadowed Rice (SR) channel, which is widely used in satellite communication systems. Besides, all the other links in the system are assumed to experience Nakagami-m channels. The channel gains for the links of satellite-Dave, satellite-Eve, UAV-Eve, UAV-Dave are expressed as h_{SD} , h_{SE} , h_{UE} , h_{UD} , respectively. According to the fading characteristics of the Nakagami-m channel, $|h_{SE}|^2$, $|h_{UE}|^2$, $|h_{UD}|^2$ follow the gamma distribution $\Gamma(m_0, \theta_0)$, $\Gamma(m_1, \theta_1)$, $\Gamma(m_2, \theta_2)$, respectively.

Furthermore, it is believed that UAV possesses the ideal CDI for Dave's channel gains and the noise variance at Dave σ_E^2 . In addition, we assume that UAV can only acquire limited CDI about Eve's channel gains, and it is unaware of Eve's detection threshold and noise variance σ_E^2 .

III. ANALYSIS FOR COVERT PERFORMANCE

The signal received at Dave from the j th ($j \in \{1, 2, \dots, M\}$) channel during the n th time slot when the LEO satellite transmits covert message can be described as

$$\begin{aligned} y_D[j, n] = & \sqrt{\frac{P_S[n]}{d_{SD}^\alpha[n]}} h_{SD}[n] x_S[j, n] + n_D[j, n] \\ & + \sqrt{\frac{P_U[n]}{d_{UD}^\alpha[n]}} h_{UD}[n] x_U[j, n], \end{aligned} \quad (4)$$

where the j th transmit symbol of the LEO satellite and the UAV's jamming signal in the n th time slot, respectively, are represented by $x_S[j, n]$ and $x_U[j, n]$. $P_S[n]$, $P_U[n]$ represent the transmit power of the satellite and UAV, and $n_D[j, n]$ is the AWGN at Dave with the variance σ_D^2 . Therefore, Dave's covert rate in the n th time slot can be determined by

$$R[n] = \frac{1}{J} \sum_{j=1}^J R_c[j, n] \\ = \log \left(1 + \frac{\frac{P_S[n]}{d_{SD}^\alpha[n]} |h_{SD}[n]|^2}{\frac{P_U[n]}{d_{UD}^\alpha[n]} |h_{UD}[n]|^2 + \sigma_D^2} \right), \quad (5)$$

where $R_c[j, n]$ is the covert rate at Dave for the j th channel in the n th time slot and J is the total number of signals transmitted by the satellite. Meanwhile, Eve needs to determine whether the LEO satellite has sent a message to Dave in order to detect the presence of covert communication. The detection method can be transformed into a binary hypothesis test, as follows:

$$H_0 : y_E[j, n] = x_U[j, n]h_{UE}[n] \sqrt{\frac{P_U[n]}{d_{UE}^\alpha[n]}} + n_E[j, n], \quad (6)$$

$$H_1 : y_E[j, n] = x_U[j, n]h_{UE}[n] \sqrt{\frac{P_U[n]}{d_{UE}^\alpha[n]}} + n_E[j, n] \\ + x_S[j, n]h_{SE}[n] \sqrt{\frac{P_S[n]}{d_{SE}^\alpha[n]}}. \quad (7)$$

Above, the signal received at Eve over the j th channel during the n th time slot is shown as $y_E[j, n]$. Note that $n_E[j, n]$ is the AWGN with variance $\sigma_E^2[n]$, and H_1 (or H_0) specifies the hypothesis that the LEO satellite transmits covert messages or not. Moreover, the radiometer detection for this hypothesis is applied, resulting in the following decision

$$Y[n] = \frac{1}{J} \sum_{j=1}^J |y_E[j, n]|^2 \underset{D_0}{\overset{D_1}{\gtrless}} \tau[n], \quad (8)$$

where D_0 and D_1 represent decisions that support hypotheses H_0 and H_1 , and $\tau[n]$ is the pre-determined threshold. Furthermore, $Y[n]$ denotes the received signal power at Eve in n th time slot, which can be expressed as

$$H_0 : Y[n] = \sigma_E^2 + |h_{UE}[n]|^2 \frac{P_U[n]}{d_{UE}^\alpha[n]}, \quad (9)$$

$$H_1 : Y[n] = |h_{SE}[n]|^2 \frac{P_S[n]}{d_{SE}^\alpha[n]} + \sigma_E^2 + |h_{UE}[n]|^2 \frac{P_U[n]}{d_{UE}^\alpha[n]}. \quad (10)$$

Based on above analysis, let us define the false alarm probability $P_{FA} = P_r(D_1|H_0)[n]$, and the missed detection probability $P_{MD} = P_r(D_0|H_1)[n]$. Hence, the probability of detection errors $\lambda[n]$ becomes

$$\lambda[n] = P_{FA}[n] + P_{MD}[n], \quad (11)$$

Lemma 1: The false-alarm and missed-detection probabilities in n th time slot are, respectively, expressed as follows

$$P_{FA}[n] = P_r[a|h_{UE}[n]|^2 + \sigma_E^2 > \tau] \\ = \begin{cases} 1, & \tau[n] < \sigma_E^2, \\ e^{-\frac{\tau[n] - \sigma_E^2}{a}}, & \tau[n] \geq \sigma_E^2, \end{cases} \quad (12)$$

$$P_{MD}[n] = P_r[a|h_{UE}[n]|^2 + b|h_{SE}[n]|^2 + \sigma_E^2 < \tau[n]] \\ = \begin{cases} 0, & \tau[n] < \sigma_E^2, \\ \frac{a(-e^{-Q/a}) + a + b(e^{-Q/b} - 1)}{a - b}, & \tau[n] \geq \sigma_E^2, \end{cases} \quad (13)$$

where $a = \frac{P_U[n]}{d_{UE}^\alpha[n]}$, $b = \frac{P_S[n]}{d_{SE}^\alpha[n]}$, $Q = \tau[n] - \sigma_E^2$.

Proof: According to Eq. (8), it's true that $P(D_1|H_0)[n] = P(|h_{UE}[n]|^2 > \frac{Q}{a})$ and $P(D_0|H_1)[n] = P(a|h_{UE}[n]|^2 + b|h_{SE}[n]|^2 < Q)$. Let us assume that $m_0 = m_1 = \theta_0 = \theta_1 = 1$, which gives the probability density functions (PDFs) $f_{|h_{UE}[n]|^2}(x) = f_{|h_{SE}[n]|^2}(x) = e^{-x}$. Then, upon substituting the PDFs, we can derive the results in Eq. (12) and Eq. (13). Hence, Eve's probability of detection errors can be given by

$$\lambda[n] = \begin{cases} 1, & \tau[n] < \sigma_E^2 \\ e^{-Q/a} + \frac{a(-e^{-Q/a}) + a + b(e^{-Q/b} - 1)}{a - b}, & \tau[n] \geq \sigma_E^2 \end{cases}. \quad (14)$$

In practice, the UAV can't determine $\lambda[n]$ because it is unknown the threshold $\tau[n]$ which Eve adopted. Therefore, as suggested by [8], we consider the worst case for ourselves from the perspective of UAV, which means Eve has the ideal detection threshold $\tau[n]^*$, and leads to the minimal probability of detection errors $\lambda[n]^*$. ■

Theorem 1: Eve's optimal detection threshold $\tau[n]^*$ is

$$\tau[n]^* = \begin{cases} \text{nonexistent}, & \tau[n] < \sigma_E^2, \\ \frac{ab \ln ab^{-1}}{a-b} + \sigma_E^2, & \tau[n] \geq \sigma_E^2, \end{cases} \quad (15)$$

when $\tau[n] \geq \sigma_E^2$, we can get the minimum probability of detection errors for Eve as

$$\lambda[n]^* = e^{-\frac{b \ln ab^{-1}}{a-b}} + \frac{-ae^{-\frac{b \ln ab^{-1}}{a-b}} + a - b + be^{-\frac{a \ln ab^{-1}}{a-b}}}{a - b}. \quad (16)$$

Proof: Eq.(14) shows the detection error probability $\lambda[n]$ of Eve, in order to get $\lambda[n]^*$, we need to find the optimal decision threshold $\tau[n]^*$.

a): When $\tau[n] < \sigma_E^2$, we can't get the minimized λ because its value is 1, it can be seen that τ does not exist.

b): In this case, we find out the first derivative of $\lambda[n]$ with respect to $\tau[n]$ as $\frac{\partial \lambda[n]}{\partial \tau[n]} = \frac{1}{b-a} e^{-\frac{Q}{b}} - \frac{b}{a(b-a)} e^{-\frac{Q}{a}}$, when $\frac{\partial \lambda[n]}{\partial \tau[n]} = 0$, we can get $\tau[n]^* = \frac{ab \ln ab^{-1}}{a-b} + \sigma_E^2$, then $\lambda[n]^*$ can be shown in Eq. (16). ■

IV. PROBLEM FORMULATION FOR UAV JAMMING

In terms of the covert transmission of LEO sat communications, we intend to maximize the covert rate $R[n]$ at Dave, as well as to maximize $\lambda[n]$ at Eve by jointly optimizing UAV's flight speed $v[n]$, flight angle $\omega_0[n]$, $\omega_1[n]$ and its

jamming power $P_U[n]$. Thus, it is possible to formulate the multi-objective optimization problem as

$$\begin{aligned}
& (\mathcal{P}_1) \max_{\{v[n], P_U[n], \omega_0[n], \omega_1[n]\}} R[n], \\
& \text{and } (\mathcal{P}_2) \max_{\{v[n], P_U[n], \omega_0[n], \omega_1[n]\}} \lambda[n], \quad (17) \\
& \text{s.t.} \quad 0 \leq v[n] \leq v_{max}, \quad (a) \\
& \quad 0 \leq P_U[n] \leq P_U^{max}, \quad (b) \\
& \quad 0^\circ \leq \omega_0[n], \omega_1[n] \leq 360^\circ, \quad (c) \\
& \quad z_U[n] \geq H_{min}, \quad (d) \\
& \quad S \leq L_{max}^2, \quad (e)
\end{aligned}$$

where the maximum flight speed and transmit power for the UAV are constrained by the variables v_{max} and P_U^{max} , respectively. At the same time, the UAV's flying height can't be less than H_{min} in order to prevent the presence of interference by the enemy Eve. In addition, the UAV's flight distance in one time slot, which is $(x_U[n] - x_U[n-1])^2 + (y_U[n] - y_U[n-1])^2 + (z_U[n] - z_U[n-1])^2$, is represented by S . The maximum distance L_{max} symbolizes the UAV's ability to fly at the fastest speed throughout time slot n is equal to $v_{max} \times \delta_t$.

Evidently, typical optimization techniques are unable to effectively address the multi-objective problem posed above. First, because this problem is non-convex, conventional optimization techniques have a hard time solving it. Second, UAV needs to know Eve's exact detection threshold $\tau[n]$ to solve $(\mathcal{P}_2)$, while the information was not provided according to the practical system considerations. Third, the UAV should be able to access the immediate channel information associated to Eve, while in this work only the partial CDI of the channels are available.

Therefore, we convert the goal of $(\mathcal{P}_2)$ to obtain the minimum probability of detection errors $\lambda[n]^*$ which satisfy the higher constraint (f) to achieve the purpose of $\max \lambda$. The system's next optimization problem can be characterized as

$$\begin{aligned}
& (\mathcal{P}_1) \max_{\{v[n], P_U[n], \omega_0[n], \omega_1[n]\}} R[n], \quad (18) \\
& \text{s.t.} \quad \lambda[n]^* \geq 1 - \varepsilon, \quad (f) \\
& \quad (a), (b), (c), (d), (e),
\end{aligned}$$

we use GA to resolve the single-objective problem after that.

V. GA BASED UAV JAMMING OPTIMIZATION ALGORITHM

On the basis of Darwin's theory of the survival of the fittest, GA were proposed in 1975. It can exit local search area and enter global search space to find the best solutions [9]. And it has been proved to have powerful search ability both in theory and practice and has important practical significance in practical engineering applications.

In this section, we'll apply GA to solve the optimization problem in Eq.(18). We assume that the UAV flies continuously in the communication process. GA algorithm consists of four steps, which are described in detail as follows

A. Initial Population

Each iteration in GA is referred to as a generation, and each generation has m entities and the population of the l th generation $G^{m,l} = \{p^{0,l}, \dots, p^{m,l}\}$ represents the set of entities, where $p^{i,l}$ represents the i th entity, $i \in (0, \dots, m)$. In this article, each entity represents a solution to optimization problem $(\mathcal{P}_1)$, containing four genes which are $v, P_U, \omega_0, \omega_1$, respectively. We need to initialize the population at first, i.e., the population $G^{m,0}$ should be created as the initial solution participate in the next iteration.

B. Evaluate Fitness and Selection

The fitness function, also known as the evaluation function, is a criterion for distinguishing entities in a population. In this paper, according to the optimization goal, we set the fitness function to Eq.(5). In the process of each iteration, we need to calculate the fitness score of $p^{i,l}$, and then, sort these entities according to their scores, select them with the higher scores as parents to produce new individuals. We adopted the strategy of stochastic tournament.

C. Crossover

Crossover refers to the exchange of some genes between two parent entities of the same generation in a certain way, thus forming two new offspring individuals. We use One-point Crossover to exchange partial values from them.

D. Mutation

The population's variety can be maintained during the mutation phase, and it can also address the problem of the algorithm failing to find a local optimal solution, and prevent the algorithm from convergence prematurely. Mutated individuals randomly change their genes to produce new offspring. We use the bit flip mutation operator to perform mutation operations.

Repeat step B to D until the algorithm converges. After continuous iterations and population evolution, and the algorithm converges, we can get the optimal individual of the population as the optimal solution to problem $(\mathcal{P}_1)$, which is also considered to be the jamming scheme of the UAV next time slot. And then the UAV will interfere with Eve according to the solution, so as to realize covert communication of the LEO satellite.

VI. SIMULATION RESULTS

We present the simulation results for the problem under consideration in this section. Let us predefine the coordinates of the legitimate user Dave on the ground are (100, 1000, 0), and the coordinates of Eve are (150, 1200, 0). In addition, $H_{min} = 30m$, $p_U^{max} = 30dBm$, $v_{max} = 5m/s$, $\sigma_D = -30dBm$, $\alpha = 2$, $\delta_t = 1s$, $T = 100s$. The take-off position of the UAV is set to (232, 1060, 120). Besides, we assume that $P_S = 40dBm$, $h_{SD} = 1.2$, $h_{UD} = 0.9$, $d_{SD} = d_{SE} = 130km$ and antenna gain G is 20dbi. We set different constraints on λ^* , taking values of 0.1, 0.05, 0.02 for ε , i.e. $\lambda^* \geq 0.9$, $\lambda^* \geq 0.95$, $\lambda^* \geq 0.98$,

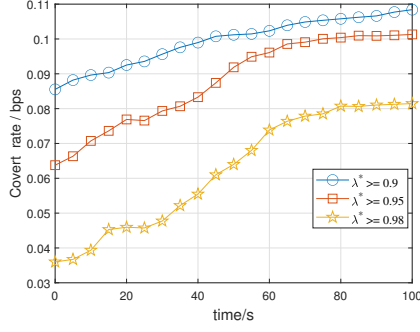


Figure 2. Covert rate R comparison

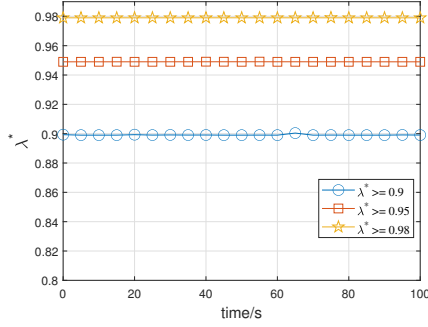


Figure 3. Minimal probability of detection errors λ^* of eavesdropper comparison

In Fig.2, we investigate the change in covert rate R during the communication process. It shows that R increases as UAV's flight time T increases, which means that the whole communication process is optimized towards the goal of ($\mathcal{P}_1$). We can also observe that when λ^* increases, the value of R decreases, which illustrates that the improvement of λ^* obtained at the cost of sacrificing R .

Fig.3 clearly shows the value of the minimum probability of detection errors λ^* under different constraints, which reveals that the UAV's jamming solution generated by GA algorithm can make the behavior of covert communication difficult to detect, and also proves the effectiveness of GA algorithm.

Fig.4 evaluates UAV's jamming power versus the flight period T . It shows that p_U decreases as T increases, this is because, as the UAV is constantly approaching the eavesdropper, its power consumption can be reduced. Further, we should notice that the UAV can only make the interference more intense by maintaining higher power, thus making the λ^* higher for eavesdroppers.

VII. CONCLUSION

In this paper, we have studied the optimization of UAV's trajectory and power under the LEO communication system, based on the premise that the UAV only knows part of the environmental information and does not know the detection threshold of eavesdroppers. The multi-objective optimization problem is formulated to maximize the covert

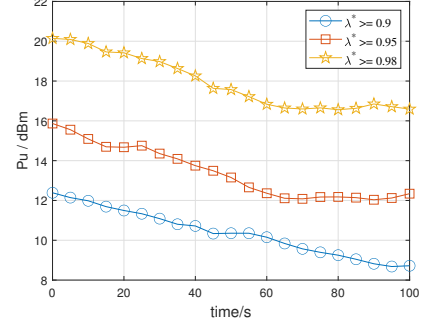


Figure 4. UAV's transmit power p_U comparison

rate and the probability of detection errors of this model using the UAV's jamming technique. This problem can be converted to a single-objective problem and solved by genetic algorithm (GA). The simulation results demonstrate that the GA-generated UAV jamming solution can significantly enhance the LEO systems' covert performance. In addition, GA only requires a smaller amount of knowledge than other benchmark algorithms, indicating that GA is a promising candidate for covert communication systems.

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